Partial Differential Equations

PENG

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1 Introduction

2 ODE

Definition 2.1 (smooth)**.** *Smooth means that a function is continuously differentiable.*

Definition 2.2 (ODE)**.** *An ordinary differential equation is an equation of the form [1]*

$$\dot{x} = f(t, x, \lambda) \tag{2.1}$$

Example 2.3 (initial-value problem)**.** *An known function $u(t)$ of the single independent variable t $(0 < t < \infty)$ using data about its derivative $u'(t)$, where the value of u at the initial time $t = 0$ is available [5].*

$$u'(t) = Cu(t) \tag{2.2}$$

with initial condition

$$u(0) = u_0 \tag{2.3}$$

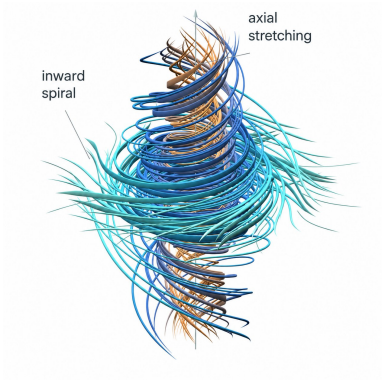


Figure 1.1: FINITE TIME BLOWUP FOR NAVIER-STOKES [4].

The solution of the initial-value problem is

$$u(t) = e^{Ct}u_0 \tag{2.4}$$

2.1 Stability

Definition 2.4 (stable ODE)**.** *Stability is an important mathematical property of the ODE, which requires no explicit solution.*

- *a stable solution to the ODE is a solution that remains bounded as t increases*
- *an unstable solution grows indefinitely as t increases*

Example 2.5. *Assume S is a complex number of the form*

$$C = \Re(C) + i\Im(C) \tag{2.5}$$

where $i = \sqrt{-1}$ is the imaginary number, and $\Re(C)$ and $\Im(C)$ are the real and imaginary parts of C, respectively.

$$u(t) = u_0 e^{t\Re(C)} e^{it\Im(C)} \tag{2.6}$$

The solution $u(t)$ is stable if and only if $\Re(C) \leq 0$.

Definition 2.6 (vector ODE)**.** *$u(t)$ is a n-dimensional vector, with individual components that are scalar functions of t.*

$$u(t) = \begin{bmatrix} u_1(t) \\ \vdots \\ u_n(t) \end{bmatrix} \tag{2.7}$$

$$u'(t) = \begin{bmatrix} u_1'(t) \\ \vdots \\ u_n'(t) \end{bmatrix} \tag{2.8}$$

$$u_0 = \begin{bmatrix} u_{01} \\ \vdots \\ u_{0n} \end{bmatrix} \tag{2.9}$$

The solution of n-dimensional vector ODE takes the form

$$u(t) = e^{Ct}u_0 = \begin{bmatrix} u_{01}e^{C_1 t} \\ \vdots \\ u_{0n}e^{C_n t} \end{bmatrix} \tag{2.10}$$

where e^{Ct} is $n \times n$ matrix

$$e^{Ct} = \begin{bmatrix} e^{C_1 t} & 0 & \dots & 0 \\ 0 & e^{C_2 t} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & e^{C_n t} \end{bmatrix} \tag{2.11}$$

Definition 2.7 (mesh of discrete time)**.**

$$t = h, 2h, \ldots, jh, \ldots \tag{2.12}$$

where h is a fixed and small parameter of meshsize.

$$u(jh) = e^{jCh}u_0 = e^{Ch}e^{(j-1)Ch}u_0 = e^{Ch}u((j-1)h) \tag{2.13}$$

2.2 Stability in vector ODE

Definition 2.8 (stable vector ODE)**.** *A system of ODEs is called stable if it has not unstable (unbounded) solution.*

Let C be the Jordan form

$$C = J^{-1}\Lambda J \tag{2.14}$$

2.3 Stable Invariant Subspace

Definition 2.9 (stable invariant subspace)**.**

2.4 Nonhomogeneous Case

Definition 2.10 (nonhomogeneous ODE)**.**

$$u'(t) = Cu(t) + f(t) \tag{2.15}$$

where f is a given function that is integrable in $[0, \infty)$

$$\int_0^\infty |f(\tau)|d\tau < \infty \tag{2.16}$$

The solution to the initial-value problem is [1, Chapter 4]

$$u(t) = e^{Ct}u_0 + \int_0^t f(\tau)d\tau \tag{2.17}$$

3 PDE

3.1 Notation

Definition 3.1 (n-dimensional real Euclidean space)**.** $\mathbb{R}^n$

Definition 3.2 (point in $\mathbb{R}^n$)**.**

$$x := \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \tag{3.1}$$

Definition 3.3 ($u(x)$)**.**

$$u : U \to \mathbb{R} \Leftrightarrow u(x) := u(x_1, \ldots, x_n) \quad (x \in U) \tag{3.2}$$

Definition 3.4 (standard coordinate vector)**.**

$$e_i := \begin{bmatrix} 0 \\ \vdots \\ 0 \\ 1 \\ \vdots \\ 0 \end{bmatrix} \tag{3.3}$$

Definition 3.5 (derivative)**.**

$$u_{x_i} := \frac{\partial u}{\partial x_i}(x) = \lim_{h \to 0} \frac{u(x + he_i) - u(x)}{h} \tag{3.4}$$

Definition 3.6 (smooth)**.** *u is smooth provided u is* ***infinitely differentiable****.*

Definition 3.7 (multi-index of order)**.**

$$\alpha = \begin{bmatrix} \alpha_1 \\ \vdots \\ \alpha_n \end{bmatrix} \tag{3.5}$$

$$|\alpha| = \alpha_1 + \ldots + \alpha_n \tag{3.6}$$

where $\alpha_i \geq 0$

Definition 3.8 (D^α)**.**

$$D^\alpha u(x) := \frac{\partial^{|\alpha|} u(x)}{\partial x_1^{\alpha_1} \ldots \partial x_n^{\alpha_n}} = \partial_{x_1}^{\alpha_1} \ldots \partial_{x_n}^{\alpha_n} u \tag{3.7}$$

Definition 3.9 (D^k)**.**

$$D^k u(x) := \{D^\alpha u(x) : |\alpha| = k\} \tag{3.8}$$

Definition 3.10 (gradient vector)**.**

$$Du = \nabla u := \begin{bmatrix} \frac{\partial u}{\partial x_1} \\ \vdots \\ \frac{\partial u}{\partial x_n} \end{bmatrix} \in \mathbb{R}^n \tag{3.9}$$

Definition 3.11 (Hessian matrix)**.** *Real symmetrical matrices:*

$$D^2 u := \begin{bmatrix} u_{x_1 x_1} & \ldots & u_{x_1 x_n} \\ \vdots & \ddots & \vdots \\ u_{x_n x_1} & \ldots & u_{x_n x_n} \end{bmatrix} \tag{3.10}$$

Definition 3.12 ($A : B$)**.**

$$A : B = \sum_{i=1}^{m} \sum_{j=1}^{n} a_{ij} b_{ij} \tag{3.11}$$

Definition 3.13 ($|A|$)**.**

$$|A| = \sqrt{\sum_{i=1}^{m} \sum_{j=1}^{n} a_{ij}^2} \tag{3.12}$$

Definition 3.14 (vector dot product and matrix multiplication)**.**

$$A \cdot B = A^T B = [a_1 \ldots a_n] \begin{bmatrix} b_1 \\ \vdots \\ b_n \end{bmatrix} = \sum_{i=1}^{n} a_i b_i \tag{3.13}$$

Definition 3.15 (radial derivative)**.**

$$\begin{aligned} u_r :=& \frac{x}{|x|} \cdot Du = \frac{1}{|x|} [x_1 \ldots x_n] \begin{bmatrix} \frac{\partial u}{\partial x_1} \\ \vdots \\ \frac{\partial u}{\partial x_n} \end{bmatrix} \\ =& \frac{\sum_{i=1}^{n} x_i \frac{\partial u}{\partial x_i}}{\sqrt{\sum_{i=1}^{n} x_i^2}} \quad (x \neq 0) \end{aligned} \tag{3.14}$$

Definition 3.16 ($\boldsymbol{u}$)**.**

$$\boldsymbol{u} : U \to \mathbb{R}^m \Leftrightarrow \boldsymbol{u} := \begin{bmatrix} u^1 \\ \vdots \\ u^m \end{bmatrix} \tag{3.15}$$

Definition 3.17 ($D^\alpha \boldsymbol{u}$)**.**

$$D^\alpha \boldsymbol{u} := \begin{bmatrix} D^\alpha u^1 \\ \vdots \\ D^\alpha u^n \end{bmatrix} \tag{3.16}$$

Definition 3.18 ($D^k \boldsymbol{u}$)**.**

$$D^k \boldsymbol{u} := \{D^\alpha \boldsymbol{u} : |\alpha| = k\} \tag{3.17}$$

Definition 3.19 (gradient matrix)**.**

$$D\boldsymbol{u} := \begin{bmatrix} u^1_{x_1} & \cdots & u^1_{x_n} \\ \vdots & \ddots & \vdots \\ u^m_{x_1} & \cdots & u^m_{x_n} \end{bmatrix} \tag{3.18}$$

3.2 PDE

Definition 3.20 (k^{th}-order PDE)**.**

$$F(D^k u(x), D^{k-1} u(x), \ldots, Du(x), u(x), x) = 0 \tag{3.19}$$

where $x \in U$, $u : U \to \mathbb{R}$, *and* $F : \mathbb{R}^{n^k} \times \mathbb{R}^{n^{k-1}} \times \cdots \times \mathbb{R}^n \times \mathbb{R} \times U \to \mathbb{R}$

Definition 3.21 (linear PDE)**.**

$$\sum_{|\alpha| \le k} a_\alpha(x) D^\alpha u = f(x) \tag{3.20}$$

$f \equiv 0 \Rightarrow$ *homogeneous*

Definition 3.22 (semi-linear)**.**

$$\sum_{|\alpha| = k} a_\alpha(x) D^\alpha u + a_0(D^{k-1} u, \ldots, Du, u, x) = 0 \tag{3.21}$$

Definition 3.23 (quasi-linear)**.**

$$\sum_{|\alpha| = k} a_\alpha(D^{k-1} u, \ldots, Du, u, x) D^\alpha u + a_0(D^{k-1} u, \ldots, Du, u, x) = 0 \tag{3.22}$$

Remark 3.24 (nonlinear)**.** *Full non-linearity depends on the highest order derivatives.*

Definition 3.25 (k^{th}-order system of PDE)**.**

$$F(D^k \boldsymbol{u}(x), D^{k-1} \boldsymbol{u}(x), \ldots, D\boldsymbol{u}(x), \boldsymbol{u}(x), x) = 0 \tag{3.23}$$

where $x \in U$, $\boldsymbol{u} : U \to \mathbb{R}^m$, *and* $F : \mathbb{R}^{mn^k} \times \mathbb{R}^{mn^{k-1}} \times \cdots \times \mathbb{R}^{mn} \times \mathbb{R}^m \times U \to \mathbb{R}^m$

3.3 Solutions to PDE

Definition 3.26 (solutions to PDE [2])**.**

- *classic solutions*
  - *explicit, closed-form formula: polynomials, exp, cos, sin*
  - *or at least show such a solution exists, and to deduce various of its properties*
- *weak solutions and regularity (smoothness)*

Remark 3.27. *Existence is the prime attribute of what we study.*

4 Four Fundamental Linear PDE

Definition 4.1 (Fundamental linear PDE)**.**

- *Transport equation*
- *Laplace's equation*
- *Heat equation*
- *Wave equation*

Lemma 4.2. $\frac{dy}{d(x+b)} = \frac{dy}{dx}$

Proof.

$$\frac{dy}{d(x+b)} = \frac{\frac{dy}{dx}}{\frac{d(x+b)}{dx}} = \frac{dy}{dx}$$

□

4.1 Transport Equation

Definition 4.3 (Transport equation)**.**

$$u_t + b \cdot Du = \frac{\partial u}{\partial t} + [b_1 b_2 \dots b_n] \begin{bmatrix} \frac{\partial u}{\partial x_1} \\ \frac{\partial u}{\partial x_2} \\ \vdots \\ \frac{\partial u}{\partial x_n} \end{bmatrix} = \frac{\partial u}{\partial t} + \sum_{i=n}^{n} b_i \frac{\partial u}{\partial x_i} = 0 \tag{4.1}$$

Example 4.4. *For 1D and 2D cases:*

$$\frac{\partial u}{\partial t} + b\frac{\partial u}{\partial x} = 0$$
$$\frac{\partial u}{\partial t} + b_1\frac{\partial u}{\partial x} + b_2\frac{\partial u}{\partial y} = 0$$

Theorem 4.5. *For Eq. 4.1, u is constant on the line $(x,t) + (b,1)s$ for $s \in \mathbb{R}$.*

Proof. Let $z(s) := u(x + bs, t + s)$.

$$\frac{dz}{ds} = \frac{\partial u}{\partial(t+s)}\frac{d(t+s)}{ds} + \frac{\partial u}{\partial(x+sb)}\frac{d(x+sb)}{ds} = \frac{\partial u}{\partial t} + b\frac{\partial u}{\partial x} \overset{\text{Eq. } 4.1}{=} 0$$
$$z(s) = u(x + sb, t + s) = C$$

□

Proposition 4.6 (initial-value problem)**.**

$$\begin{cases} \frac{\partial u}{\partial t} + b\frac{\partial u}{\partial x} = 0 \\ u(x,0) = g(x) \end{cases}$$

$$u(x,t) \overset{Theo.\ 4.5}{=} u(x - tb, 0) = g(x - tb)$$

Proposition 4.7 (nonhomogenous problem)**.**

$$\begin{cases} \frac{\partial u}{\partial t} + b \cdot \frac{\partial u}{\partial x} = f(x,t) \\ u(x,0) = g(x) \end{cases}$$

Let $z(s) := u(x + sb, t + s)$.

$$\frac{dz}{ds} = f(x+sb, t+s)$$

$$z(s) = u(x+sb, t+s) = \int f(x+sb, t+s)ds$$

$$z(0) - z(-t) = u(x,t) - u(x-tb, 0) = u(x,t) - g(x-tb) = \int_{-t}^{0} f(x+sb, t+s)ds$$

$$u(x,t) = g(x-tb) + \int_{-t}^{0} f(x+sb, t+s)ds$$

4.2 Laplace's Equation

Definition 4.8 (Laplace's equation)**.**

$$\Delta u = \nabla \cdot \nabla u = \nabla^2 u = \left[\frac{\partial}{\partial x_1}\frac{\partial}{\partial x_2}\cdots\frac{\partial}{\partial x_n}\right]\begin{bmatrix}\frac{\partial u}{\partial x_1}\\ \frac{\partial u}{\partial x_2}\\ \vdots \\ \frac{\partial u}{\partial x_n}\end{bmatrix} = \sum_{i=1}^{n}\frac{\partial^2 u}{\partial x_i^2} = 0 \tag{4.2}$$

Example 4.9. *For 1D and 2D cases:*

$$\frac{\partial^2 u}{\partial x^2} = 0$$

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

Definition 4.10 (outward derivative)**.**

$$\frac{\partial u}{\partial v} := \boldsymbol{v} \cdot Du = [v_1 v_2 \dots v_n]\begin{bmatrix}\frac{\partial u}{\partial x_1}\\ \frac{\partial u}{\partial x_2}\\ \vdots \\ \frac{\partial u}{\partial x_n}\end{bmatrix} = \sum_{i=1}^{n} v_i \frac{\partial u}{\partial x_i}$$

where $\boldsymbol{v}$ is the unit outer normal vector of ∂U.

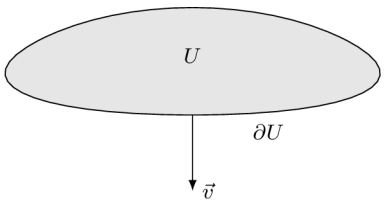


Figure 4.1: Outward derivative.

Example 4.11. *For 1D and 2D cases:*

$$\frac{\partial u}{\partial v} = v\frac{\partial u}{\partial x}$$

$$\frac{\partial u}{\partial v} = v_1\frac{\partial u}{\partial x} + v_2\frac{\partial u}{\partial y}$$

Theorem 4.12 (Gauss-Green theorem)**.**

$$\begin{aligned}\int_U \nabla \cdot \boldsymbol{u} dx &= \int_U \begin{bmatrix}\frac{\partial}{\partial x_1}\frac{\partial}{\partial x_2}\dots\frac{\partial}{\partial x_n}\end{bmatrix}\begin{bmatrix}u_1\\u_2\\\vdots\\u_n\end{bmatrix} dx = \int_U \sum_{i=1}^n \frac{\partial u_i}{\partial x_i} dx \\ = \int_{\partial U} \boldsymbol{u}\cdot\boldsymbol{v} dS &= \int_{\partial U} [u_1 u_2 \dots u_3]\begin{bmatrix}v_1\\v_2\\\vdots\\v_n\end{bmatrix} dS = \int_{\partial U}\sum_{i=1}^n u_i v_i dS\end{aligned} \tag{4.3}$$

Example 4.13. *For 1D and 2D cases:*

$$\int_U \frac{\partial u}{\partial x} dx = \int_{\partial U} uv dS$$
$$\int_U \frac{\partial u_1}{\partial x} + \frac{\partial u_2}{\partial y} dx = \int_{\partial U} u_1 v_1 + u_2 v_2 dS$$

Fundamental solution:

$$u(x) = v(r)$$
$$r = |x| = \sqrt{\sum_{i=1}^n x_i^2}$$
$$\frac{\partial r}{\partial x_i} = \frac{x_i}{r}$$
$$\frac{\partial u}{\partial x_i} = \frac{dv}{dr}\frac{x_i}{r}$$
$$\frac{\partial^2 u}{\partial x_i^2} = \frac{d^2 v}{dr^2}\frac{x_i^2}{r^2} + \frac{dv}{dr}\frac{r^2 - x_i^2}{r^3}$$
$$\Delta u = \sum_{i=1}^n \frac{\partial^2 u}{\partial x_i^2} = \frac{d^2 v}{dr^2} + \frac{dv}{dr}\frac{n-1}{r} = 0$$

For $n = 1$:

$$\frac{d^2 v}{dr^2} = 0$$

$$v(r) = r + C$$

For $n \geq 2$:

$$\frac{d^2 v}{dr^2} + \frac{dv}{dr}\frac{n-1}{r} = 0$$

$$\frac{v''(r)}{v'(r)} = \ln\left(|v'(r)|\right)' = \frac{1-n}{r}$$

$$\ln\left(|v'(r)|\right) = (1-n)\ln(r) + \ln(C) = \ln\left(\frac{C}{r^{n-1}}\right)$$

$$|v'(r)| = \frac{C}{r^{n-1}}$$

$$v'(r) = \frac{C}{r^{n-1}}$$

For $n = 2$:

$$v'(r) = \frac{C}{r}$$

$$v(r) = C_1 \ln(r) + C_2$$

For $n \geq 3$:

$$v(r) = \frac{C_1}{2-n} \frac{1}{r^{n-2}} + C_2 = \frac{C_1}{r^{n-2}} + C_2$$

Definition 4.14 (fundamental solution of Laplace's equation)**.**

$$\Phi(|x|) := \begin{cases} -\frac{1}{2\pi} \ln |x| & (n=2) \\ \frac{1}{n(n-2)\alpha(n)} \frac{1}{|x|^{n-2}} & (n \geq 3) \end{cases} \tag{4.4}$$

Definition 4.15 (Poisson's equation)**.**

$$-\Delta u = -\sum_{i=1}^{n} \frac{\partial^2 u}{\partial x_i^2} = f \tag{4.5}$$

Theorem 4.16. *Solution of Poisson's equation:*

$$u(x) = \int_{y \in \mathbb{R}^n} \Phi(|x-y|) f(y) dy = \begin{cases} -\frac{1}{2\pi} \int_{\mathbb{R}^n} \ln(|x-y|) f(y) dy & (n=2) \\ \frac{1}{n(n-2)\alpha(n)} \int_{\mathbb{R}^n} \frac{f(y)}{|x-y|^{n-2}} dy & (n \geq 3) \end{cases} \tag{4.6}$$

5 Numerical Methods

5.1 ODE

5.1.1 Numerical Solution

Definition 5.1 (numerical scheme)**.** *Numerical solution of an ODE is a numerical approximation to the solution $u(t)$ of the original ODE on a grid of discrete time $t_1, t_2, \ldots$.*

6 Asymptotic Methods

7 Analytical Methods

A Tools

Definition A.1 (Euler's formula [3])**.**

$$e^{ix} = \cos(x) + i \sin(x) \tag{A.1}$$

A.1 Taylor Expansion

Definition A.2 (Taylor expansion)**.**

A.2 Exponent

Definition A.3 (exponent)**.** *The exponent function e^x is defined as an infinite series*

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots = \sum_{n=0}^{\infty} \frac{x^n}{n!} \tag{A.2}$$

Approximate the exponent function by the truncated Taylor polynomial

$$T_K(x) = \sum_{n=0}^{K} \frac{x^n}{n!} \tag{A.3}$$

which is a good approximation to e^x when x is rather small (i.e. $|x| < 0.5$).

When x is large, pick a sufficiently large integer m such that $x/2^m$ is sufficiently small [5]

$$e^x = (e^{x/2^m})^{2^m} \tag{A.4}$$

Approximate by

$$e^x \doteq T_K(x/2^m)^{2^m} \tag{A.5}$$

Definition A.4 (Horner's algorithm)**.**

Definition A.5 (Pade approximation)**.**

Definition A.6 (exponent of matrix)**.**

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